

**BOARD QUESTION PAPER:
MARCH 2014 Physics**

Time: 3 Hours

Total Marks: 70

Note:

- i. All questions are compulsory.
- ii. Answers to the two sections are to be written in the same answer book.
- iii. Figures to the right hand side indicate full marks.
- iv. Draw neat and labelled diagrams, wherever necessary.
- v. Use of logarithmic table is allowed.
- vi. Answer to every question must be started on a new page.

SECTION – I

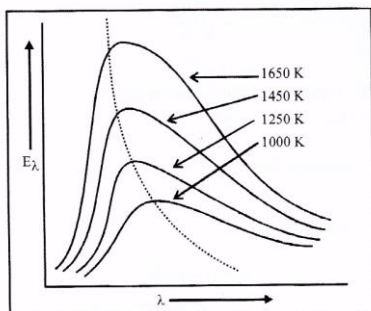
Q.1. Attempt any six: [12]

- (i) Explain the rise of liquid in the capillary on the basis of pressure difference. [2]
- (ii) Show graphical representation of energy distribution spectrum of perfectly black body. [2]
- (iii) The escape velocity of a body from the surface of the earth is 11.2 km/s. If a satellite were to orbit close to the surface, what would be its critical velocity? [2]
- (iv) A pipe which is open at both ends is 47 cm long and has an inner diameter 5 cm. If the speed of sound in air is 348 m/s, calculate the fundamental frequency of air column in that pipe. [2]
- (v) Show that R.M.S. velocity of gas molecules is directly proportional to square root of its absolute temperature. [2]
- (vi) For a particle performing uniform circular motion $\vec{v} = \vec{\omega} \times \vec{r}$. Obtain an expression for linear acceleration of the particle performing non- uniform circular motion. [2]
- (vii) A stone of mass 1 kg is whirled in horizontal circle attached at the end of a 1 m long string. If the string makes an angle of 30° with vertical, calculate the centripetal force acting on the stone. ($g = 9.8 \text{ m/s}^2$). [2]
- (viii) A solid cylinder of uniform density of radius 2 cm has mass of 50 g. If its length is 12 cm, calculate its moment of inertia about an axis passing through its centre and perpendicular to its length. [2]

Solution 1:

(i) When a capillary tube is dipped in water, the water meniscus inside the tube is concave. The pressure just below the meniscus is less than the pressure just above it by $2T/R$, where T is the surface tension of water and R is the radius of curvature of meniscus. The pressure on the surface of water is atmospheric pressure P . The pressure just below the plane surface of water outside the tube is also P , but that just below the meniscus inside the tube is $P - (2T/R)$. We know that pressure at all points in the same level of water must be same. Therefore, to make up the deficiency of pressure, $2T/R$, below the meniscus, water begins to flow from outside to inside the tube. The rising of water in the capillary takes place.

(ii)



(iii)

$$V_e = \sqrt{2}v_c$$

$$\Rightarrow v_c = \frac{11.2}{\sqrt{2}}$$

$$\Rightarrow v_c = 7.91 \text{ km/s}$$

(iv) For open pipe,

Given:

$L = 47 \text{ cm}$, inner diameter $= 5 \text{ cm}$ and $v = 348 \text{ m/s}$

Fundamental frequency, $n = ?$

Since,

$$n = \frac{v}{2L}$$

For pipe open at both ends, corrected length of air column,

$$L = l + 2e$$

$$= l + 2(0.3d)$$

$$= l + 0.6d$$

Thus,

$$n = \frac{v}{2(l + 0.6d)}$$

$$\Rightarrow n = \frac{348}{2(0.47 + 0.6 \times 5 \times 10^{-2})}$$

$$\Rightarrow n = 348 \text{ Hz}$$

(v) We know that,

$$P = \frac{1}{3} \frac{mn}{V} \times v^2$$

$$\Rightarrow PV = \frac{1}{3} M \times v^2$$

But, $PV = RT$ (by Gas equation)

Thus,

$$v^2 = \frac{3RT}{M}$$

But,

$$v_{rms} = \sqrt{v^2}$$

$$= \sqrt{\frac{3RT}{M}}$$

Hence, $v_{rms} \propto \sqrt{T}$

(vi) Let the particle be moving around a circular path of constant radius r . If it speeds up or slows down, its motion is non-uniform and both the angular speed (ω) and the linear speed (v) change with time. Thus, at any instant, ω , v and r as related as:

$$\vec{v} = \vec{\omega} \times \vec{r}$$

On differentiating, we get,

$$\frac{d\vec{v}}{dt} = \frac{d(\vec{\omega} \times \vec{r})}{dt}$$

$$\Rightarrow \frac{d\vec{v}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

$$\text{But } \frac{d\vec{\omega}}{dt} = \vec{\alpha} \text{ and } \frac{d\vec{r}}{dt} = \vec{v}$$

$$\therefore \frac{d\vec{v}}{dt} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

$$\vec{a} = \vec{a}_1 + \vec{a}_2$$

Where, $\vec{a}_1$ = tangential component

$\vec{a}_2$ = Radial component

(vii) Given: $m = 1 \text{ kg}$, $r = 1 \text{ m}$, $\theta = 30^\circ$

$$\text{Now, Centripetal force} = \frac{mv^2}{r}$$

$$\text{Thus, Centripetal force} = \frac{1 \times v^2}{r}$$

And

$$\begin{aligned} v &= \sqrt{rg \tan \theta} \\ &= \sqrt{1 \times 9.8 \times \tan 30^\circ} \\ &= 2.37 \text{ m/s} \end{aligned}$$

Again, Centripetal force

$$\begin{aligned} &= \frac{mv^2}{r} \\ &= \frac{1 \times (2.37)^2}{1} \\ &= 5.61 \text{ N} \end{aligned}$$

(viii) Given $R = 2 \text{ cm}$, $M = 50 \text{ gm}$, $L = 12 \text{ cm}$

$$\begin{aligned} I_{\text{cm}} &= M \left(\frac{R^2}{4} + \frac{L^2}{12} \right) \\ &= 50 \times 10^{-3} \left[\frac{(2 \times 10^{-2})^2}{4} + \frac{(12 \times 10^{-2})^2}{12} \right] \\ \text{Now,} \\ &= 50 \times 10^{-3} \times [0.0001 + 0.0012] \\ &= 5 \times 10^{-2} \times 13 \times 10^{-4} \\ &= 65 \times 10^{-6} \\ &= 6.50 \times 10^{-5} \text{ kgm}^2 \end{aligned}$$

Question 2: Attempt any THREE:

[9]

(i) Derive an expression for acceleration due to gravity at depth 'd' below the earth's surface. [3]

(ii) A copper metal cube has each side of length 1 m. The bottom edge of the cube is fixed and tangential force $4.2 \times 10^8 \text{ N}$ is applied to a top surface. Calculate the lateral displacement of the top surface if modulus of rigidity of copper is $14 \times 10^{10} \text{ N/m}^2$. [3]

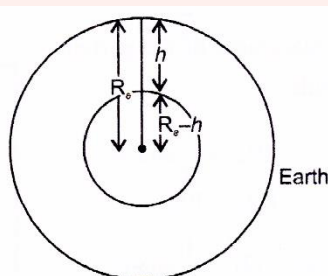
(iii) State an expression for K.E. (kinetic energy) and P.E. (potential energy) at displacement 'x' for a particle performing linear S.H.M. Represent them graphically. Find the displacement at which K.E. is equal to P.E. [3]

(iv) The equation of simple harmonic progressive wave is given by $Y = 0.05 \sin \pi \left[20t - \frac{x}{6} \right]$

Where all quantities are in S.I. units. Calculate the displacement of a particle at 5m from origin and at the instant 0.1 second. [3]

Solution 2:

(i) Let a body P of mass m be situated at a depth h below the earth's surface.



The gravitational force of attraction on a body inside a spherical shell is always zero. Therefore, the body P experiences gravitational attraction only due to the inner solid sphere. The mass of this sphere is

$$M' = \text{volume} \times \text{density} = \frac{4}{3} \pi (R_e - h)^3 \rho$$

Where, ρ is mean density of the earth. Therefore, according to Newton's law of gravitation, the force of attraction on the body P is

$$\begin{aligned} \frac{GM'm}{(R_e - h)^2} &= G \frac{\frac{4}{3} \pi (R_e - h)^3 \rho m}{(R_e - h)^2} \\ &= \frac{4}{3} \pi G (R_e - h) \rho m \end{aligned}$$

This force must be equal to the weight of the body mg' , where g' is the acceleration due to gravity at a depth h below the surface of the earth. Thus,

$$mg' = \frac{4}{3} \pi G (R_e - h) \rho m \text{ -----(i)}$$

Similarly, if a body be at the surface of the earth ($h = 0$) where acceleration due to gravity is g then,

$$mg = \frac{4}{3} \pi G R_e \rho m \text{ ----- (ii)}$$

Now, dividing equation (i) and (ii), we get,

$$\frac{g'}{g} = \frac{R_e - h}{R_e}$$

$$\Rightarrow g' = g \left(1 - \frac{h}{R_e} \right)$$

This is the expression for acceleration due to gravity at depth h below the earth surface.

(ii) Given: Length, $l = h = 1\text{m}$, $F = 4.2 \times 10^8 \text{ N}$, $\eta = 14 \times 10^{10} \text{ N / m}^2$

Now,

$$\eta = \frac{Fh}{Ax}$$

$$\Rightarrow 14 \times 10^{10} = \frac{4.2 \times 10^8 \times 1}{1 \times 1 \times x}$$

$$\Rightarrow x = \frac{4.2 \times 10^8}{14 \times 10^{10}}$$

$$= 0.003\text{m}$$

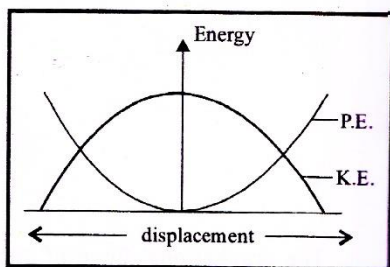
$$= 3 \times 10^{-3}\text{m}$$

$$\text{(iii) K.E.} = \frac{1}{2} m \omega^2 (a^2 - y^2)$$

Where, y is the displacement of the particle and a is the amplitude.

And

$$\text{P.E} = \frac{1}{2} m \omega^2 y^2$$



Now, we have,

$$\text{P.E.} = \text{K.E.}$$

Then,

$$\begin{aligned}\frac{1}{2} m \omega^2 (a^2 - y^2) &= \frac{1}{2} m \omega^2 y^2 \\ \Rightarrow \frac{1}{2} m \omega^2 a^2 - \frac{1}{2} m \omega^2 y^2 &= \frac{1}{2} m \omega^2 y^2 \\ \Rightarrow \frac{1}{2} m \omega^2 a^2 &= \frac{1}{2} m \omega^2 y^2 + \frac{1}{2} m \omega^2 y^2 \\ \Rightarrow \frac{1}{2} m \omega^2 a^2 &= m \omega^2 y^2 \\ \Rightarrow y &= \pm \frac{a}{\sqrt{2}}\end{aligned}$$

(iv) Given: $x = 5\text{cm}$, $t = \text{sec}$

$$Y = 0.05 \sin \pi \left[20t - \frac{x}{6} \right]$$

Thus, displacement,

$$\begin{aligned}Y &= 0.05 \sin \pi \left[20t - \frac{x}{6} \right] \\ &= 0.05 \sin \pi (2 - 0.833) \\ &= -0.025\text{m}\end{aligned}$$

Question 3:

[7]

State and prove the theorem of 'parallel axes'.

Calculate the density of paraffin oil, if glass capillary of diameter 0.25 mm dipped in paraffin oil of surface tension 0.0245 N/m rises to a height of 4 cm.

(Angle of contact of paraffin with glass = 28° and acceleration due to gravity = 9.8 m/s^2 .)

OR

A wire of density ' ρ ' and Young's modulus ' Y ' is stretched between two rigid supports separated by a distance ' L ' under tension ' T '. Derive an expression for its frequency in fundamental mode.

Hence show that $n = \frac{1}{2L} \sqrt{\frac{YI}{\rho L}}$, where symbols have their usual meanings.

When the length of a simple pendulum is decreased by 20 cm, the period changes by 10%. Find the original length of the pendulum.

Solution 3:

Theorem of parallel axis:- It states that the moment of inertia of a body about any axis equal to the sum of the moment of inertia of parallel axis through its center of mass and the product of their masses and the square of its perpendicular distance between them.

Proof:- Let us consider a rigid body of mass ' M ' rotating about an axis passing through at ' o '.

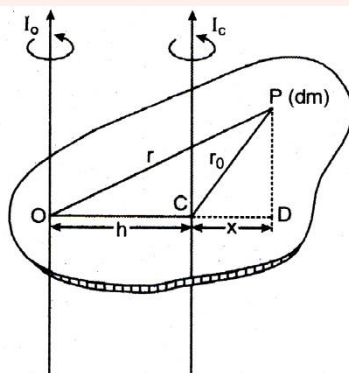
Suppose I_o is the moment of inertia of rigid body about an axis passing through o .

Also, consider another axis parallel through first axis passing through point c and perpendicular to the plane.

I_c is the moment of inertia of the body about the second axis.

Let $oc = b$ be the distance between two parallel axes

Consider an element having mass ' dm ' situated at the point p .



Therefore,

$$I_0 = \int r^2 dm$$

$$= \int r^2 dm \dots\dots(1)$$

$$\therefore I_c = \int Cp^2 dm$$

$$= \int r_0^2 dm \quad (2)$$

Now, draw PD perpendicular to OC.

So,

$$OP^2 = OD^2 + PD^2$$

$$= (h + CD)^2 + PD^2$$

$$= h^2 + 2hCD + CD^2 + PD^2$$

$$= h^2 + 2hCD + CP^2$$

$$\therefore OP^2 = CP^2 + h^2 + 2hCD$$

Now, multiply by 'dm' in both sides and integrating both sides

$$\int r^2 dm = \int r_0^2 dm + \int h^2 dm + \int 2hxdm$$

$$\Rightarrow \int r^2 dm = \int r_0^2 dm + h^2 \int dm + 2h \int xdm \dots(3)$$

Since,

$$\int xdm = 0$$

As the algebraic sum of moments of all particles.

Therefore, for a body in equilibrium

$$\int dm = M$$

From eq(1),(2) and (3)

$$I_0 = I_c + Mn^2$$

Numerical:

Density of Paraffin oil = ρ

Diameter of capillary = 0.25mm; $r = 0.125 \times 10^{-3} \text{ m}$

Surface Tension, $T = 0.0245 \text{ N/m}$

Height, $h = 4 \text{ cm}$

$G = 98 \text{ m/s}^2$

$\theta = 28^\circ$

Since,

$$T = \frac{r\rho g}{2 \cos \theta}$$

$$\Rightarrow 0.0245 = \frac{0.125 \times 10^{-3} \times 4 \times 10^{-2} \times \rho \times 9.8}{2 \times \cos 28^\circ}$$

$$\Rightarrow \rho = 882 \text{ kg/m}^3$$

OR

Consider a wire stretched between two rigid supports at a distance L apart. Let

T = the tension in the wire

R = the radius of cross section of the wire

Y, ρ = young's modulus and mass density of the material of the wire.

M, m = the mass and linear density of the wire

Then, $M = (\pi r^2 L) \rho$ and $m = \frac{M}{L} = \pi r^2 \rho$ ----- (i)

Since, The stress in the wire = $\frac{T}{\pi r^2}$

Thus, $\frac{T}{m} = \frac{T}{\pi r^2 \rho} = \frac{\text{stress}}{\rho}$ ----- (ii)

The fundamental frequency of vibration of the wire,

$$n = \frac{1}{2L} \sqrt{\frac{T}{m}}$$

$$= \frac{1}{2L} \sqrt{\frac{\text{stress}}{\rho}} \text{ ----- (iii)}$$

If $\Delta L = l$ is the elastic extension of the wire under tension T, Strain = l/L

Since, $Y = \frac{\text{Stress}}{\text{Strain}}$

$$\Rightarrow \text{Stress} = Y \times \text{strain} = Y \frac{l}{L} \text{ ----- (iv)}$$

Thus,

$$n = \frac{1}{2L} \sqrt{\frac{Yl}{\rho L}} \text{ ----- (v)}$$

Which is the required expression.

Numerical:

Let original length = l

Length decreased by 20 cm and period changes by 10%.

Since,

$$T = 2\pi\sqrt{\frac{l}{g}} \text{----- (i)}$$

And new time period will decreased by 10%

$$= T - \frac{10}{100} \times T$$

$$= T - \frac{T}{10} = \frac{9T}{10}$$

$$\Rightarrow \frac{9T}{10} = 2\pi\sqrt{\frac{l-20}{g}} \text{----- (ii)}$$

$$\Rightarrow \frac{T}{\frac{9T}{10}} = \frac{2\pi\sqrt{\frac{l}{g}}}{2\pi\sqrt{\frac{l-20}{g}}}$$

$$\Rightarrow \frac{10}{9} = \sqrt{\frac{l}{l-20}}$$

$$\Rightarrow \left(\frac{10}{9}\right)^2 = \frac{l}{l-20}$$

$$\Rightarrow \frac{100}{81} = \frac{l}{l-20}$$

Now on further simplification we get,

$$\Rightarrow 100(l-20) = 81l$$

$$\Rightarrow 100l - 81l = 2000$$

$$\Rightarrow 19l = 2000$$

$$\Rightarrow l = \frac{2000}{19} \text{ cm} = \frac{20}{19} \text{ m}$$

Hence, $l = 1.05\text{m}$

Question 4. Select and write the most appropriate answer from the given alternatives for each sub-question: [7]

(i) The bulging of earth at the equator and flattening at the poles is due to _____.

- (a) Centripetal force (b) centrifugal force (c) gravitational force (d) electrostatic force

(ii) Young's modulus of material of wire is 'Y' and strain energy per unit volume is 'E', then the strain is

- (a) $\sqrt{\frac{Y}{2E}}$ (b) $\sqrt{\frac{E}{Y}}$ (c) $\sqrt{\frac{2E}{Y}}$ (d) $\sqrt{2EY}$

(iii) The wavelength range of thermal radiation is

- (a) From 4000 Å to 7000 Å (b) from 7700 Å to 4×10^6 Å
(c) From 10^6 Å to 10^8 Å (d) from 4×10^{-12} Å to 4×10^8 Å

(iv) A pipe open at both ends resonates to a frequency 'n₁' and a pipe closed at one end resonates to a frequency 'n₂'. If they are joined to form a pipe closed at one end, then the fundamental frequency will be _____.

- (a) $\frac{n_1 n_2}{2n_2 + n_1}$ (b) $\frac{2n_2 n_1}{2n_2 + n_1}$ (c) $\frac{2n_2 n_1}{n_1 + n_2}$ (d) $\frac{n_2 + 2n_1}{n_2 n_1}$

(v) The phase difference between displacement and acceleration of a particle performing S.H.M. is _____.

- (a) $\frac{\pi}{2}$ rad (b) π rad (c) 2π rad (d) $\frac{3\pi}{2}$ rad

(vi) Let n₁ and n₂ be the two slightly different frequencies of two sound waves. The time interval between waxing and immediate next waning is _____.

- (a) $\frac{1}{n_1 - n_2}$ (b) $\frac{2}{n_1 - n_2}$ (c) $\frac{n_1 - n_2}{2}$ (d) $\frac{1}{2(n_1 - n_2)}$

(vii) A metal ball cools from 64 °C to 50 °C in 10 minutes and to 42 °C in next 10 minutes. The ratio of rates of fall of temperature during the two intervals is _____.

- (a) $\frac{4}{7}$ (b) $\frac{7}{4}$ (c) 2 (d) 2.5

Solution 4:

(i) (b) centrifugal force

(ii) (c) $\sqrt{\frac{2E}{Y}}$

(iii) (b) from 7700 Å to 4×10^6 Å

(iv) (a) $\frac{n_1 n_2}{2n_2 + n_1}$

(v) (b) π rad

(vi) (d) $\frac{1}{2(n_1 - n_2)}$

(vii) (b) $\frac{7}{4}$

Section –II

Question 5: Attempt any SIX:

[12]

(i) Show that the orbital magnetic dipole moment of a revolving electron is $\frac{evr}{2}$.

(ii) Describe the construction of photoelectric cell.

(iii) For a glass plate as a polariser with refractive index 1.633, calculate the angle of incidence at which light is polarised.

(iv) The susceptibility of magnesium at 300 K is 2.4×10^{-5} . At what temperature will the susceptibility increase to 3.6×10^{-5} ?

(v) Draw a neat labelled diagram for Davisson and Germer experiment, for diffraction of electron wave.

(vi) Explain the terms : (a) Transmitter and (b) receiver in communication system.

(vii) A metal rod $\frac{1}{\sqrt{\pi}}$ m long rotates about one of its ends perpendicular to a plane whose

magnetic induction is 4×10^{-3} T. Calculate the number of revolutions made by the rod per second if the e.m.f. induced between the ends of the rod is 16 mV.

(viii) Find the wave number of a photon having energy of 2.072 eV.

Given : Charge on electron = 1.6×10^{-19} C,

Velocity of light in air = 3×10^8 m/s,

Planck's constant = 6.63×10^{-34} J-s.

Solution 5:

(i) Let r , v and T be the orbital radius, speed and period of motion of the electron.

$$\text{Then, } T = \frac{2\pi r}{v}$$

The orbital motion of the electron constitutes a loop of convectional current I in the opposite sense of its revolution.

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

So, the magnetic dipole moment associated with this electronic current loop is given by

$$\begin{aligned} M_o &= I \times 2\pi r^2 = \frac{ev}{2\pi r} \times \pi r^2 \\ &= \frac{1}{2} \times evr \\ &= \frac{evr}{2} \end{aligned}$$

(ii) Photoelectric cell consists of highly evacuated or gas filled glass tube, an emitter and a collector. The light enters through a quartz window and falls on the semi cylindrical cathode C coated with a photosensitive metal. The anode is in the form of straight wire of platinum or nickel, co-axial with the cathode. If the cell is required to respond to the visible part of the spectrum, the cathode is coated with potassium or rubidium and the quartz window is replaced by glass.

(iii) Given: $n = 1.633$

$$n = \tan i_p$$

$$\Rightarrow 1.633 = \tan i_p$$

$$\Rightarrow i_p = \tan^{-1}(1.633)$$

$$= 5805^\circ \text{C}$$

(iv) Given: $\chi_{m_1} = 2.4 \times 10^{-5}$, $T_1 = 300\text{K}$, $\chi_{m_2} = 3.6 \times 10^{-5}$, $T_2 = ?$

Since,

$$\chi_m \propto \frac{1}{T}$$

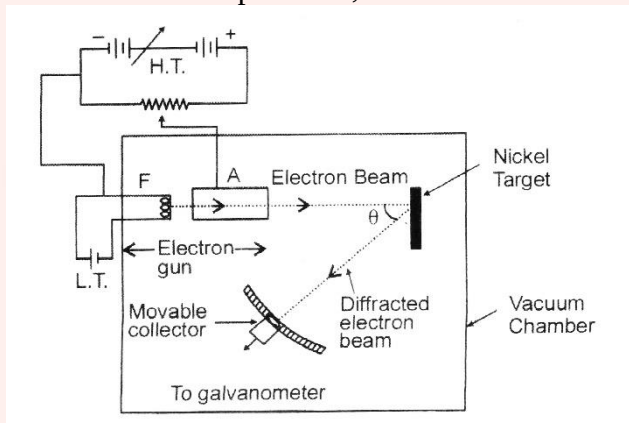
$$\chi_m \propto \frac{1}{T}$$

$$\Rightarrow \frac{\chi_{m_1}}{\chi_{m_2}} = \frac{T_2}{T_1}$$

$$\Rightarrow \frac{2.4 \times 10^{-5}}{3.6 \times 10^{-5}} = \frac{T_2}{300}$$

$$\Rightarrow T_2 = 200K$$

(v) A diagram for Davisson and Germer experiment, for diffraction of electron wave.



(vi) (a) **Transmitter:** It is a device that processes an incoming signal so as to make it suitable for transmission and then transmits it.

(b) **Receiver:** A receiver is a device that selects a signal from among all the signals received from a communication channel, recovers the base signal and delivers it to the user.

(vii)

$$E = \frac{1}{2} B \omega l^2$$

$$\Rightarrow 16 \times 10^{-3} = \frac{1}{2} \times 4 \times 10^{-3} \times \omega \times \left(\frac{1}{\sqrt{\pi}} \right)^2$$

$$\Rightarrow 16 \times 10^{-3} = \frac{1}{2} \times 4 \times 10^{-3} \times \omega \times \frac{1}{\pi}$$

$$\Rightarrow n = \frac{16 \times 10^{-3} \times 2 \times 2}{4 \times 10^{-3} \times 2 \times 2}$$

$$\Rightarrow n = 4$$

(viii)

$$E = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{hc}{E}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2.072 \times 1.6 \times 10^{-19}}$$

$$= 6 \times 10^{-7} \text{ m}$$

Now, Wave number,

$$\bar{\nu} = \frac{1}{\lambda}$$

$$= \frac{1}{6 \times 10^{-7}}$$

$$= 1.67 \times 10^6 \text{ m}^{-1}$$

Question 6. Attempt any THREE:

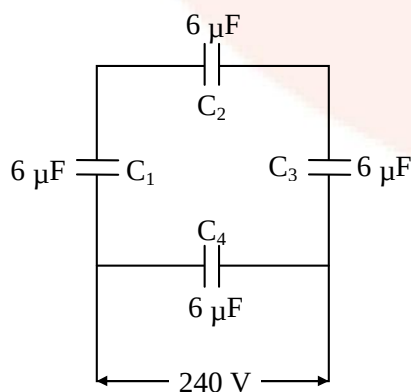
[9]

(i) State Ampere's circuital law. Obtain an expression for magnetic induction along the axis of toroid.

(ii) Calculate the radius of second Bohr orbit in hydrogen atom from the given data. Mass of electron = $9.1 \times 10^{-31} \text{ kg}$ Charge on the electron = $1.6 \times 10^{-19} \text{ C}$ Planck's constant = $6.63 \times 10^{-34} \text{ J-s}$. Permittivity of free space = $8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$

(iii) Explain the working of P-N junction diode in forward and reverse biased mode.

(iv) A network of four capacitors of $6 \mu\text{F}$ each is connected to a 240 V supply. Determine the charge on each capacitor.



Solution 6:

(i) **Ampere's circuital law:** In free space, the line integral of magnetic induction around a closed path in a magnetic field is equal to μ_0 times the net steady current enclosed by the path.

Magnetic induction along the axis of toroid:

Consider a toroidal solenoid of radius r having centre O , carrying current I .

According to Ampere's circuital law,

$$\oint \vec{B} d\vec{l} = \mu_0 I$$

Now, current I flow through the ring as many times as there are number of turns. If N is the total number of turns, then the total current flowing through toroid is NI .

$$\oint \vec{B} d\vec{l} = \mu_0 (NI) \text{ ----- (i)}$$

$\vec{B}$ and $d\vec{l}$ are in same direction

$$\oint \vec{B} d\vec{l} = B \cdot dl = B(2\pi r) \text{ ----- (ii)}$$

Comparing equation (i) and (ii), we get,

$$\mu_0 NI = B(2\pi r)$$

$$B = \frac{\mu_0 NI}{2\pi r} \text{ -----(iii)}$$

Now, if n is the number of turns per unit length of toroid, then $n = \frac{N}{2\pi r}$,

Putting the value of $n = \frac{N}{2\pi r}$ in equation (iii), we get,

$$B = \mu_0 nI$$

Which is the expression for magnetic induction along the axis of toroid.

(ii) WE know that,

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}$$

For hydrogen $Z = 1$

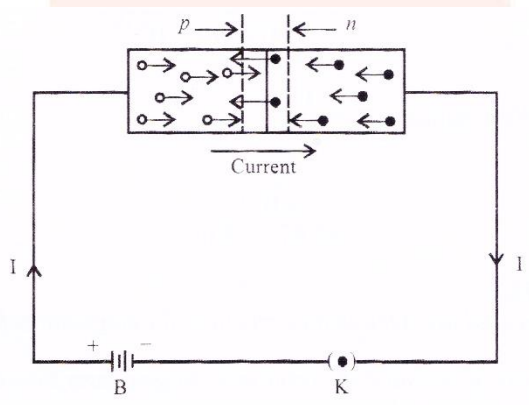
$$\text{Thus, } r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2}$$

Radius of second Bohr orbit,

$$\begin{aligned} r_2 &= \frac{2^2 \times (6.63 \times 10^{-34})^2 \times 8.85 \times 10^{-12}}{3.14 \times 9.1 \times 10^{-31} \times (1.6 \times 10^{-19})^2} \\ &= \frac{1556.07 \times 10^{-80}}{73.15 \times 10^{-69}} \\ &\approx 21.3 \times 10^{-11} \text{ m} \end{aligned}$$

(iii) **Forward- biased p-n junction diode:**

A p-n junction diode is said to be forward biased when the positive terminal of a cell or battery is connected to the p side of the junction and the negative terminal to the n side.

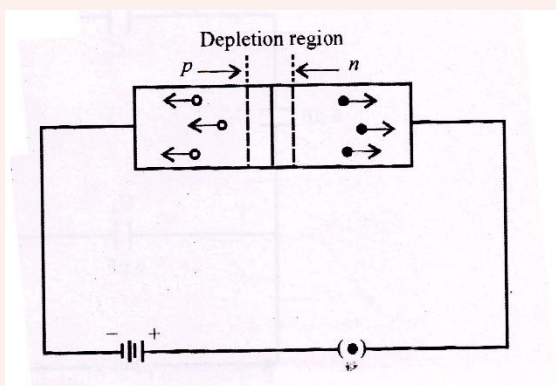


When the junction is forward-biased, the depletion region narrows and, consequently, the potential barrier is lowered. This causes the majority charge carriers of each region to cross into the other region. The electrons travel from the n- side to the p –side and go to the positive terminal of the battery. The holes that travel from p-side to n-side combine with the electrons injected into

the n- region from the negative terminal of the battery. This way the diode conducts when forward-biased.

Reverse- biased pn junction diode

A pn- junction diode is said to be reverse biased when the positive terminal of a cell or battery is connected to the n side of the junction, and the negative terminal to the p side.



With reverse bias, free electrons and holes move away from the junction. The departure of holes and electrons increase width of depletion layer and barrier potential. As there is no crossing of majority carriers across the junction, there is no current through reverse biased diode. A very small current flows because of thermally produced minority carriers in depletion region. Thus, the reverse biased diode works as an open switch.

(iv) The capacitors C_1 , C_2 and C_3 are connected in series. The equivalent capacitance C' is given by

$$\frac{1}{C'} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$

$$\Rightarrow C' = 2\mu\text{F}$$

Now, Total Capacitance, $C = 6 + 2 = 8\mu\text{F}$

In series the charge remains same. Therefore, the charge on C_1 , C_2 and C_3 is given by

$$q = C V$$

$$= 2 \times 10^{-6} \times 240$$

$$\Rightarrow q = 4.8 \times 10^{-4} \text{ coulomb}$$

The charge on capacitance C_4 is given by

$$q_4 = C_4 V$$

$$= 6 \times 10^{-6} \times 240 = 1.44 \times 10^{-3} \text{ coulomb}$$

Question 7:

[7]

Describe biprism experiment to find the wavelength of monochromatic light. Draw the necessary ray diagram for magnified and diminished images of virtual sources.

If the difference in velocities of light in glass and water is 2.7×10^7 m/s, find the velocity of light in air.

(Refractive index of glass = 1.5, Refractive index of water = 1.333)

OR

State the principle of a transformer. Explain its construction and working. Derive an expression for the ratio of e.m.f.s in terms of number of turns in primary and secondary coil.

Two diametrically opposite points of a metal ring are connected to two terminals of the left gap of Metre Bridge. The resistance of 11Ω is connected in right gap. If null point is obtained at a distance of 45 cm from the left end, find the resistance of metal ring.

Solution 7:

Distance between the source and eyepiece is directly measured on the scale marked on optical bench. And the micrometer screw fitted to eyepiece is adjusted. And to measure the fringe width β reading of corresponding micrometer is noted. The vertical wire is coincide with 5th, 10th and 15th bright fringes.

Distance d between coherent vertical sources-Conjugate foci method:

Convex lens having the focal length less than $d/4$ is mounted on the stand between eyepiece and biprism.

For position L_1 of lens:

The distance d_1 between two magnified images is measured by micrometer.

For lens L_2 :

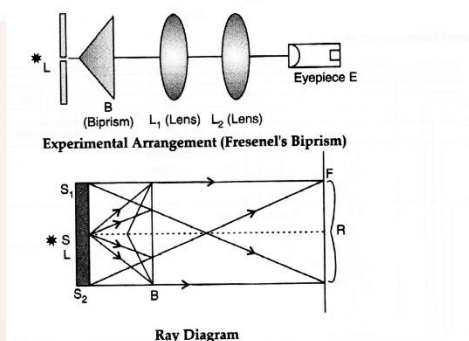
The distance d_2 diminished is measured by moving the lens towards the eyepiece.

Then, $d = \sqrt{d_1 d_2}$

Therefore, the wavelength is

$$\lambda = \frac{\beta d}{D}$$

$$= \frac{\beta \sqrt{d_1 d_2}}{D}$$



Numerical:

Given: $v_w - v_g = 2.7 \times 10^7 \text{ m/s}$, $\mu_g = 1.5$, $\mu_w = 1.333$

$$v_w = \frac{c}{\mu_w}$$

$$= \frac{c}{1.333}$$

Again,

$$v_g = \frac{c}{\mu_g}$$

$$= \frac{c}{1.5}$$

Thus,

$$v_w - v_g = \frac{c}{\mu_w} - \frac{c}{\mu_g}$$

$$= c \left[\frac{1}{1.333} - \frac{1}{1.5} \right]$$

$$\Rightarrow 2.7 \times 10^7 = c \left[\frac{1.5 - 1.333}{1.5 \times 1.333} \right]$$

$$\Rightarrow c = 323272455.1$$

$$= 3.23 \times 10^8 \text{ m/s}$$

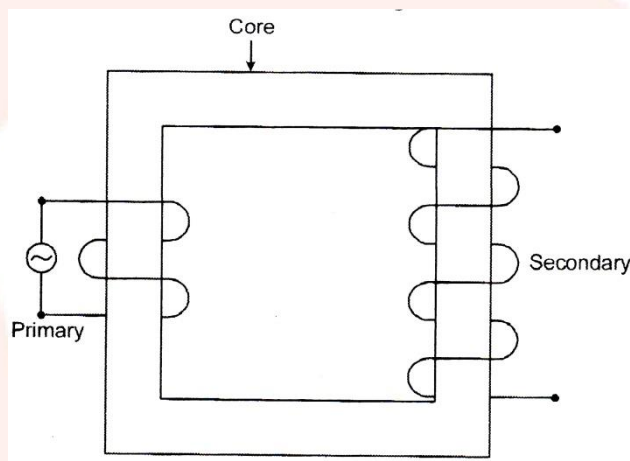
OR

The value of alternating voltage can be changed from high to low value at various stage by using a device called transformer.

Principle:- It is based on mutual induction that is production of induced emf in one coil due to change of current in neighboring coil.

Construction:-

It consists of two coils, primary (p) and secondary (s) insulated from each other and around on a soft iron core.



Primary is the input coil and secondary coil is the output coil.

Working:

- (a) When an alternating voltage is applied to the primary coil, the current through the coil goes on changing.
- (b) Magnetic flux also changes.
- (c) An emf is induced in each of two coils.

The amount of magnetic flux depends upon the number of turns of the coil.

Derivation: Let us suppose ϕ be the magnetic flux per turn linked with both the coils at certain instant 't'.

$N_p \phi$ is the magnetic flux linked with the primary coil at a certain instant 't'.

$N_s \phi$ is magnetic flux linked with the secondary coil at certain instant 't'.

Where, N_p N_s

$$\begin{aligned}\therefore e_p &= \frac{-d\phi_p}{dt} \\ &= -N_p \frac{-d\phi_p}{dt} \quad \dots(1)\end{aligned}$$

$$\begin{aligned}e_s &= \frac{-d\phi_s}{dt} \\ &= -N_s \frac{-d\phi_s}{dt} \quad \dots\dots(2)\end{aligned}$$

By taking ratio of (1) and (2)

$$\frac{e_s}{e_p} = \frac{N_s}{N_p} \quad \dots\dots(3)$$

It is known as the equation of the transformer.

Therefore, For an ideal transformer,

Input power is equal to output power,

$$\begin{aligned}e_p i_p &= e_s i_s \\ \frac{e_s}{e_p} &= \frac{i_p}{i_s} \quad \dots\dots(4)\end{aligned}$$

$\therefore$ from eq(3) and (4)

$$\therefore \frac{e_s}{e_p} = \frac{N_s}{N_p}$$

This is the ratio of emf.

Numerical:

$$R_2 = 11\Omega, l_1 = 45\text{cm}, l_2 = 100 - 45 = 55\text{cm}$$

Resistance of metal ring (R_1)=?

$$\frac{R_1}{R_2} = \frac{l_1}{l_2}$$

$$\text{Resistance of each half segment of the metal ring} = \frac{R_1}{2}$$

These half segments are connected in parallel in the left gap.

$$R_{\text{eff.}} = \frac{\frac{R_1}{2} \times \frac{R_1}{2}}{\frac{R_1}{2} + \frac{R_1}{2}} = \frac{\frac{R_1}{2} \times \frac{R_1}{2}}{R_1} = \frac{R_1^2}{4R_1}$$

$$\Rightarrow R_{\text{eff.}} = \frac{R_1}{4} \Omega$$

From the formula,

$$R_{\text{eff.}} = \frac{\frac{R_1 \times R_1}{2} \times \frac{R_1}{2}}{\frac{R_1}{2} + \frac{R_1}{2}} = \frac{\frac{R_1 \times R_1}{2} \times \frac{R_1}{2}}{R_1} = \frac{R_1^2}{4R_1}$$

$$\Rightarrow R_{\text{eff.}} = \frac{R_1}{4} \Omega$$

$$\frac{R_{\text{eff.}}}{R_2} = \frac{l_1}{l_2}$$

$$\Rightarrow \frac{\frac{R_1}{4}}{11} = \frac{45}{55}$$

$$\Rightarrow \frac{R_1}{44} = \frac{45}{55}$$

$$\Rightarrow R_1 = \frac{45}{55} \times 44$$

$$\Rightarrow R_1 = 36 \Omega$$

Hence, The resistance of metal ring is 36Ω .

Question 8: Select and write the most appropriate answer from the given alternatives for each sub-question:

[7]

(i) Intensity of electric field at a point close to and outside a charged conducting cylinder is proportional to _____. (r is the distance of a point from the axis of cylinder)

- (a) $\frac{1}{r}$ (b) $\frac{1}{r^2}$ (c) $\frac{1}{r^3}$ (d) r^3

(ii) When a hole is produced in P-type semiconductor, there is _____.

- (a) Extra electron in valence band
(b) Extra electron in conduction band
(c) missing electron in valence band
(d) missing electron in conduction band

(iii) The outermost layer of the earth's atmosphere is _____.

- (a) Stratosphere (b) mesosphere (c) troposphere
(d) ionosphere

(iv) Accuracy of potentiometer can be easily increased by _____.

- (a) increasing resistance of wire (b) decreasing resistance of wire
(c) Increasing the length of wire (d) decreasing the length of wire

(v) When electron in hydrogen atom jumps from second orbit to first orbit, the wavelength of radiation emitted is λ . When electron jumps from third orbit to first orbit, the wavelength of emitted radiation would be _____.

- (a) $\frac{27}{32}\lambda$ (b) $\frac{32}{27}\lambda$ (c) $\frac{2}{3}\lambda$ (d) $\frac{3}{2}\lambda$

(vi) An ideal voltmeter has _____.

- (a) low resistance (b) high resistance (c) infinite resistance (d) zero resistance

(vii) The resolving power of telescope of aperture 100 cm for light of wavelength 5.5×10^{-7} m is _____.

- (a) $0.149 \times 10^{+7}$ (b) $1.49 \times 10^{+7}$ (c) $14.9 \times 10^{+7}$ (d) $149 \times 10^{+7}$

Solution 8:

(i) (a) $\frac{1}{r}$

(ii) (c) missing electron in valence band

(iii) (d) ionosphere

(iv) (c) Increasing the length of wire

(v) (a) $\frac{27}{32}\lambda$

(vi) (c) infinite resistance

(vii) (a) $0.149 \times 10^{+7}$